

MiniDxO: The Transparent AI Diagnostician

- A Project to Build a "Glass Box" for Medical Reasoning

TEAM NAME :- AION

The Problem: "Black Box" AI



"DR. GOOGLE" IS OVERWHELMING,
UNRELIABLE, AND CREATES ANXIETY.



CURRENT AI MODELS OFTEN PROVIDE A
SINGLE, ONE-SHOT ANSWER ("YOU HAVE X")
WITH NO JUSTIFICATION.



LACK OF TRUST: WHEN USERS (PATIENTS OR
CLINICIANS) CAN'T SEE THE REASONING,
THEY CAN'T TRUST THE CONCLUSION.

The Solution: Introducing "MiniDxO"

A lightweight, interactive AI orchestrator that simulates a doctor's reasoning process, **MiniDxO doesn't just *give* a diagnosis; it *simulates* the diagnostic conversation.**



It achieves this through three key pillars:



Interactive Questioning: It cross-questions the user, just like a doctor.



Cited Evidence: It references trusted medical texts (e.g., Mayo Clinic, NIH) in real-time.



Total Transparency: It explains its thought process *at every step*.

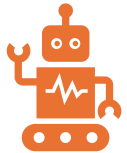
Core Concept: The Workflow)

This is how a user would experience MiniDxO.

- **AI:** What symptoms are you experiencing?
- **User:** Fever and a sore throat.
- **AI:** [Thinking...] *These symptoms are common to flu or strep throat. I need to differentiate. I will ask about a key sign of strep.*
- **AI:** Do you have white patches in your throat?
- **User:** Yes.
- **AI:** [Thinking...] *This new information strongly suggests strep. Let me verify this against a trusted source.*
- **AI: [Referencing Mayo Clinic]:** “White patches and high fever are common signs of streptococcal infection.”
- **AI:** Based on your symptoms and this information, the probable diagnosis is **Strep Throat** (Confidence: 78%).

Key Project Components

To build this, we must focus on four core components:



1. Multi-Agent Reasoning:

Not one giant AI, but a team of smaller, specialized agents
(e.g., "Symptom Triage," "Question Generator," "Source Verifier") that communicate to form a conclusion.



2. Interactive Frontend:

A simple, clean React or web-based chat interface where the reasoning is shown in real-time.



3. Medical Text Referencing:

An engine that can query, parse, and cite snippets from a pre-loaded, trusted knowledge base (MedlinePlus, NIH articles, etc.).



4. The Transparency Layer:

The "secret sauce." The UI must explicitly show the AI's *internal monologue* and *cited sources* before it gives a final answer.

Proposed Architecture (Simplified)

Knowledge Base Agent

- (Fetches & cites medical text snippets) ↓
- Orchestrator Agent** (Synthesizes all info) ↓
- Transparent UI** (Shows the final reasoning and a new question/diagnosis)

User Input ("I have a fever") ↓ **Orchestrator Agent** (The "General Practitioner") ↓
Communicates with:

Question Generator Agent

- (Decides the *best next* question to ask)

Symptom Analyzer Agent

- (Generates a list of possible conditions)

- The system's logic is based on an "Orchestrator" managing specialized agents.

Why Multi-Agent? Why Not One Big Model?

Specialization: A single model trying to do everything (chat, reason, search, cite) is complex and error-prone.



Transparency: By using separate agents, we can *literally* show their communication.



"The **Question Agent** suggests asking about 'cough' to rule out flu.



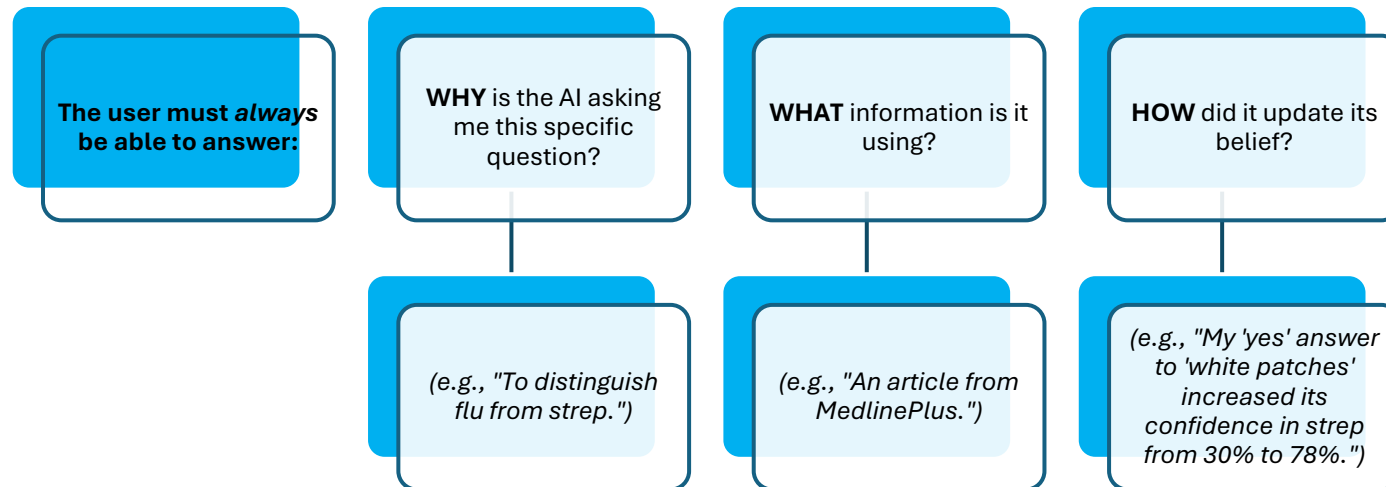
The **Knowledge Agent** confirms 'cough' is a key differentiator."



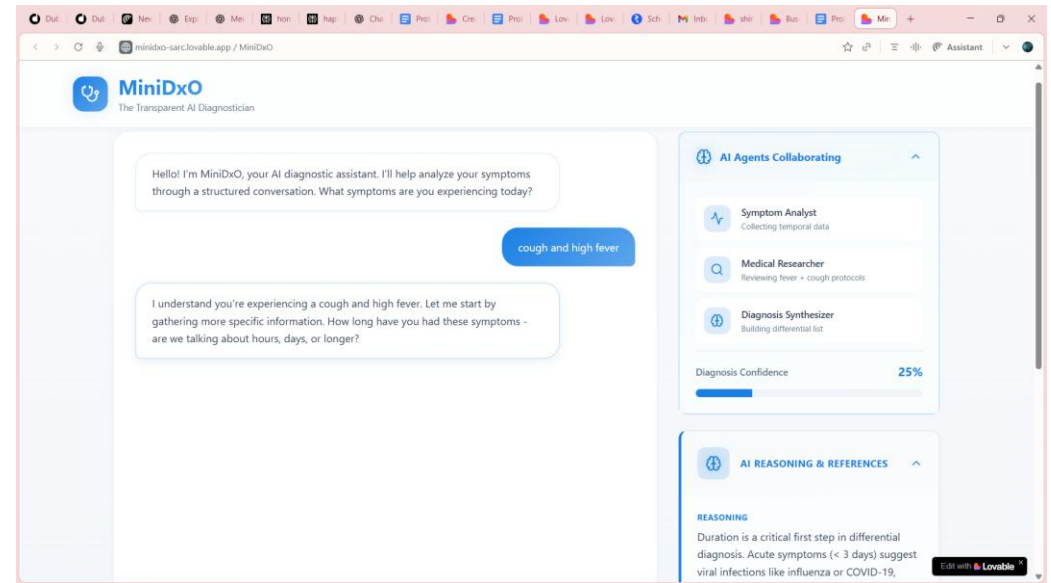
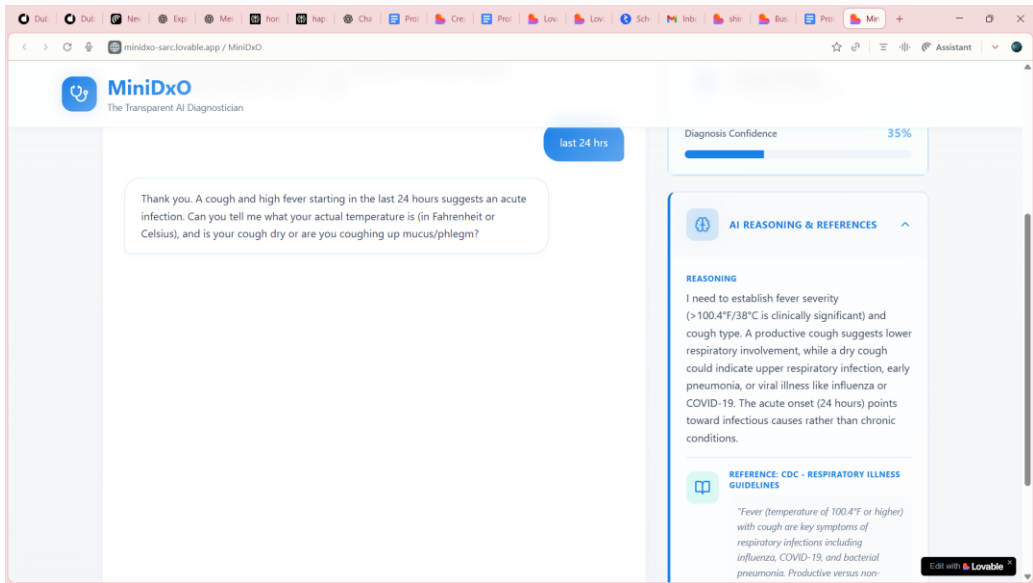
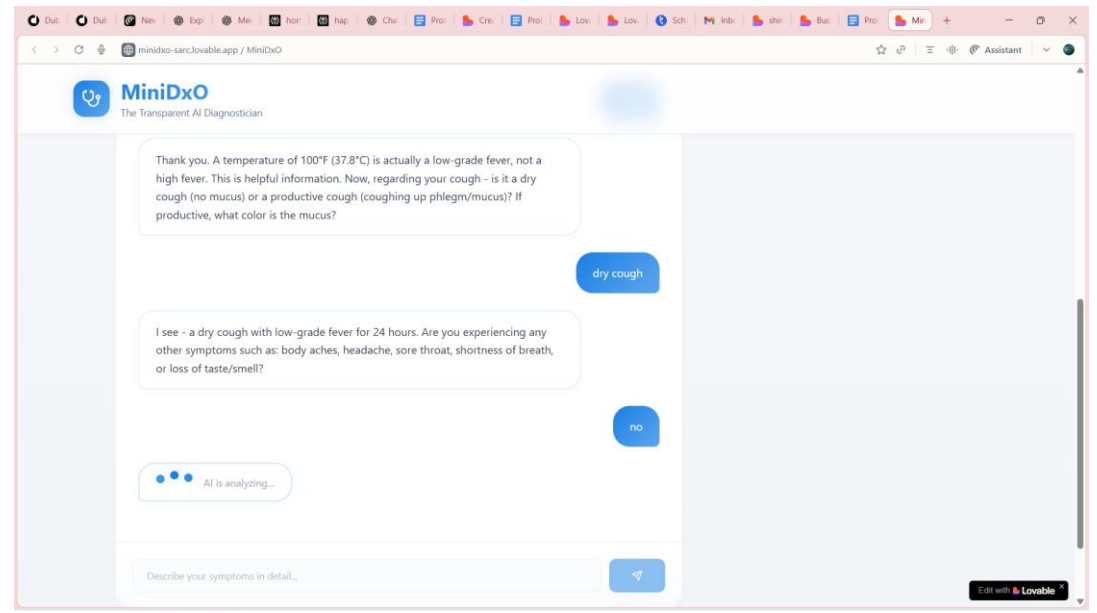
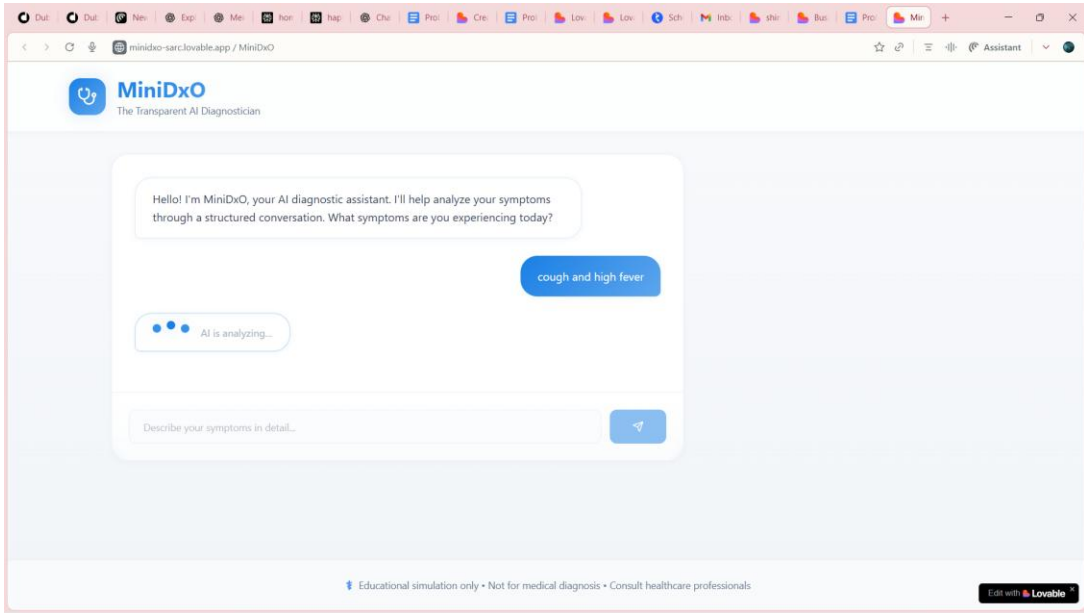
Modularity: We can swap out or upgrade one agent (like the Knowledge Base) without retraining the entire system.

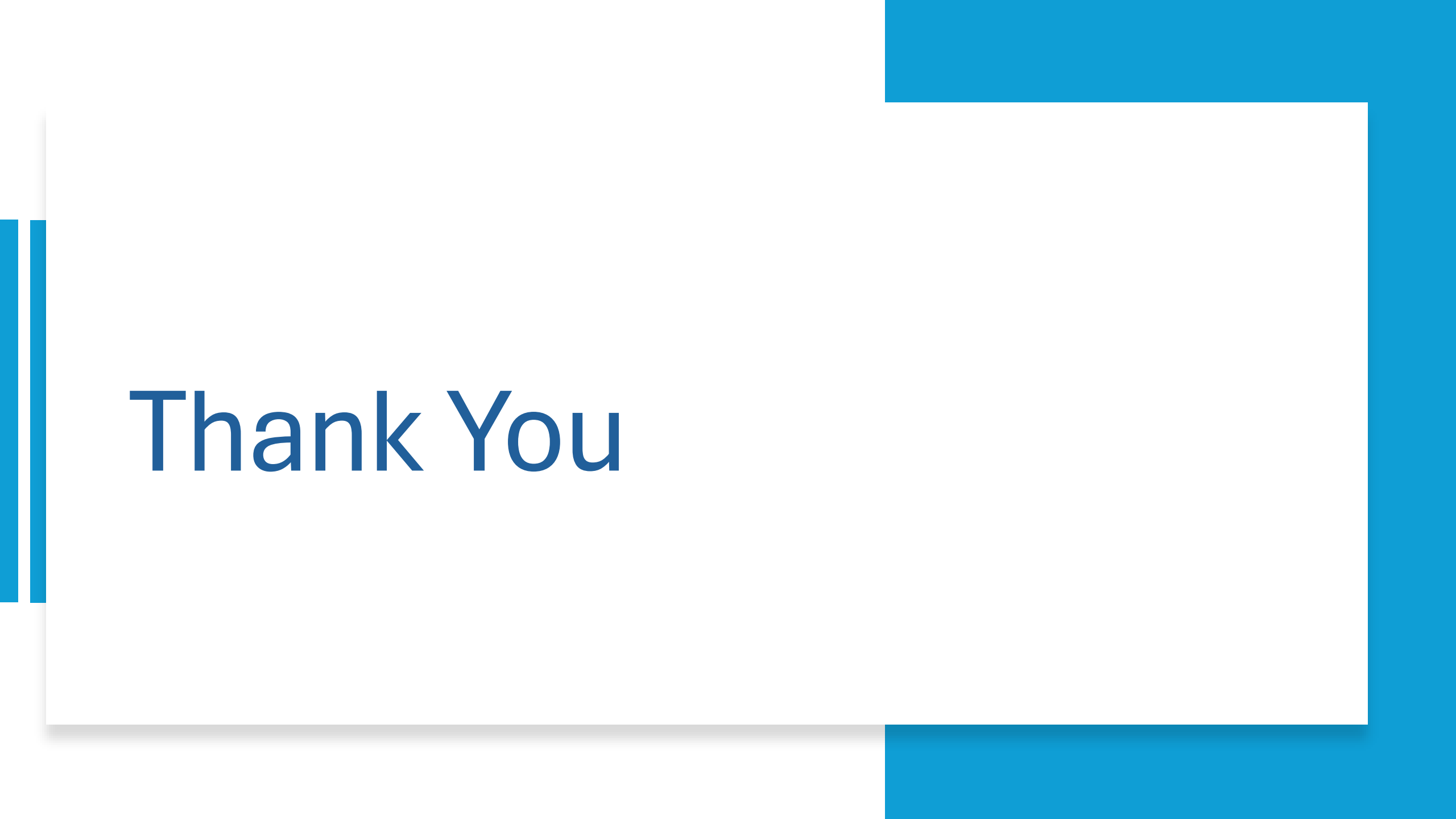
The "Transparency" Mandate

- This is the most critical design constraint.



- This builds trust and serves as a powerful educational tool.





Thank You